

HOUTRIB, Netherlands

WMO: 062680

Lat: 52.533N

Lon: 5.433E

Elev: 2

StdP: 101.30

Time Zone: 1.00 (EUC)

Period: 86-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-8.2	-5.9	-12.1	1.3	-6.9	-9.4	1.7	-5.0	18.9	7.0	17.3	7.5	5.8	70	0.746

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.6	26.2	19.4	24.2	18.4	22.6	18.1	20.9	24.6	19.8	22.9	18.7	21.5	5.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.5	14.2	23.2	18.5	13.3	21.9	17.5	12.5	20.4	60.2	24.6	56.3	22.8	52.9	21.6	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	13.4	12.2	DB	-8.4	29.8	4.7	2.3	-11.8	31.5	-14.5	32.9	-17.2	34.2	-20.6	35.9	(4)
(5)			WB	-8.8	22.9	4.7	1.1	-12.2	23.7	-15.0	24.4	-17.6	25.0	-21.1	25.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	3.5	2.3	5.9	8.4	13.0	14.8	17.8	17.3	14.4	10.7	6.4	4.4	(6)
(7)		DBStd	6.12	4.06	4.30	3.36	3.17	3.32	2.77	2.53	2.34	2.08	2.68	3.37	2.93	(7)
(8)		HDD10.0	935	203	217	131	67	7	1	0	0	0	25	112	173	(8)
(9)		HDD18.3	3109	461	450	386	297	168	111	42	48	119	238	358	431	(9)
(10)		CDD10.0	922	0	1	4	21	101	146	242	226	133	46	3	1	(10)
(11)		CDD18.3	53	0	0	0	0	4	6	26	16	1	0	0	0	(11)
(12)		CDH23.3	263	0	0	0	3	38	30	115	69	7	0	0	0	(12)
(13)		CDH26.7	43	0	0	0	0	3	3	19	17	1	0	0	0	(13)
(14)	Wind	WSAvg	6.5	7.4	6.9	7.0	6.4	6.4	6.0	6.1	6.0	5.9	6.3	6.1	6.9	(14)
(15)	Precipitation	PrecAvg	846	73	58	51	43	57	70	83	91	79	83	82	78	(15)
(16)		PrecMax	1143	142	137	101	93	96	152	154	207	179	212	151	150	(16)
(17)		PrecMin	613	8	18	11	0	27	28	27	12	13	16	9	23	(17)
(18)		PrecStd	123	34	29	28	24	21	33	38	52	40	41	35	34	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	12.4	16.5	22.1	27.0	26.4	29.3	29.0	23.6	19.3	13.6	11.8	(19)
(20)			MCWB	9.1	10.9	11.9	14.2	17.4	19.3	21.7	21.5	18.7	15.9	11.3	9.9	(20)
(21)		2%	DB	9.5	10.3	13.3	18.0	24.2	23.7	26.4	24.9	20.7	16.9	12.2	10.4	(21)
(22)			MCWB	8.2	8.9	9.8	13.6	15.9	18.1	20.2	19.3	17.1	14.7	10.7	9.1	(22)
(23)		5%	DB	8.5	8.4	11.4	15.4	21.7	21.9	24.3	22.7	18.9	15.5	11.3	9.4	(23)
(24)			MCWB	7.4	7.1	9.3	12.3	15.3	17.8	19.6	18.7	16.4	13.6	10.2	8.2	(24)
(25)		10%	DB	7.7	7.1	10.0	13.3	18.7	19.9	22.4	21.2	17.7	14.5	10.5	8.3	(25)
(26)			MCWB	6.8	6.0	8.4	10.8	14.5	16.6	18.8	18.0	15.5	12.9	9.5	7.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.0	12.0	15.6	18.3	21.0	22.7	22.5	19.4	16.2	12.0	10.1	(27)
(28)			MCDWB	10.2	12.3	15.3	20.8	23.4	24.4	27.1	28.2	22.6	18.6	12.8	11.2	(28)
(29)		2%	WB	8.3	8.9	10.7	13.8	17.1	19.5	21.3	20.5	17.9	14.9	11.1	9.2	(29)
(30)			MCDWB	9.0	10.0	12.9	17.1	22.5	22.5	25.3	24.0	19.8	16.0	11.8	10.2	(30)
(31)		5%	WB	7.6	7.2	9.5	12.5	15.8	18.1	20.2	19.2	16.9	14.1	10.4	8.4	(31)
(32)			MCDWB	8.3	8.1	11.1	15.1	20.5	20.9	23.5	22.0	18.6	15.3	11.1	9.3	(32)
(33)		10%	WB	6.9	6.1	8.6	11.1	14.7	16.6	19.0	18.3	16.0	13.1	9.6	7.5	(33)
(34)			MCDWB	7.7	6.9	9.8	13.4	18.5	19.6	21.7	20.7	17.5	14.2	10.3	8.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.5	4.3	4.9	5.8	6.4	5.4	5.6	5.8	5.0	5.1	3.7	3.2	(35)
(36)			MCDBR	3.9	5.4	7.6	9.2	11.7	9.7	9.7	9.4	7.5	6.9	4.1	4.3	(36)
(37)			MCWBR	3.6	4.4	5.2	5.7	5.8	6.0	5.6	5.3	5.0	4.9	3.6	3.9	(37)
(38)		5% WB	MCDWB	3.8	5.2	6.6	8.2	9.8	8.5	8.4	8.2	6.6	6.4	3.9	4.2	(38)
(39)			MCWBR	3.7	4.5	5.0	5.4	5.8	5.8	5.6	5.2	5.0	4.8	3.5	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.339	0.359	0.390	0.412	0.414	0.411	0.423	0.415	0.392	0.373	0.351	0.333	(40)	
(41)		taud	2.303	2.299	2.228	2.225	2.265	2.290	2.268	2.308	2.359	2.364	2.325	2.298	(41)	
(42)		Ebn at Noon	633	737	788	820	837	843	824	809	782	713	612	567	(42)	
(43)		Edh at Noon	64	86	111	125	127	125	125	114	97	80	63	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.67	1.36	2.60	4.22	5.17	5.47	5.22	4.34	3.10	1.75	0.78	0.50	(44)	
(45)		RadStd	0.07	0.24	0.28	0.39	0.31	0.34	0.52	0.34	0.26	0.15	0.07	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page